

HIGH PRAIRIE AGDM, AB, Canada

WMO: 712260

Lat: 55.3949N

Lon: 116.4831W

Elev: 1939

StdP: 13.70

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.5	-21.8	-35.3	0.8	-28.6	-28.1	1.2	-21.6	23.8	40.6	20.9	36.6	2.8	260	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	81.1	62.4	78.1	60.8	75.2	59.1	65.6	76.7	63.3	73.8	61.4	71.6	7.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	87.3	70.9	59.1	80.4	68.3	57.1	74.6	65.5	31.5	76.8	29.7	73.8	28.3	71.5	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	16.9	14.6	DB	-35.9	87.2	7.3	2.8	-41.1	89.3	-45.4	90.9	-49.5	92.5	-54.8	94.6	(4)
(5)			WB	-36.6	69.5	7.2	3.0	-41.8	71.6	-46.0	73.4	-50.1	75.0	-55.3	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	11.7	13.3	22.9	38.8	50.1	57.8	61.8	58.9	51.1	39.3	22.9	9.7
	DBStd	21.86	17.07	15.27	14.74	8.20	7.75	5.14	4.54	5.42	7.34	8.68	13.85	15.34
	HDD50	5978	1186	1027	841	343	94	5	0	4	75	342	813	1248
	HDD65	10384	1651	1447	1306	787	464	222	120	197	418	797	1262	1713
	CDD50	1104	0	0	0	7	96	238	365	281	107	9	1	0
	CDD65	35	0	0	0	0	1	5	20	9	0	0	0	0
	CDH74	863	0	0	0	7	61	144	368	208	73	2	0	0
	CDH80	129	0	0	0	1	7	13	64	32	12	0	0	0
	WSAvg	6.6	6.2	5.8	6.6	7.7	7.7	6.7	6.3	5.9	6.6	6.9	6.4	5.9
	PrecAvg	19.1	1.1	0.7	0.8	0.9	1.8	3.5	3.3	2.2	1.6	1.1	1.1	0.9
(15) Precipitation	PrecMax	25.6	2.3	1.2	1.7	1.8	3.5	6.2	5.8	4.2	4.2	2.6	2.2	2.3
	PrecMin	12.5	0.3	0.2	0.2	0.2	0.6	1.2	1.9	0.7	0.2	0.5	0.4	0.3
	PrecStd	3.1	0.5	0.3	0.3	0.4	0.7	1.3	1.1	0.8	0.7	0.5	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	46.4	53.5	72.7	80.3	81.9	85.9	83.8	81.7	70.6	57.0	46.4
		MCWB	39.2	38.4	43.0	50.3	56.3	60.5	65.6	64.8	60.9	54.2	44.4	39.2
	2%	DB	43.5	42.6	48.9	63.7	74.6	78.2	81.4	79.3	75.0	61.6	49.1	41.6
		MCWB	37.2	35.8	39.6	46.2	53.7	59.0	64.2	63.1	57.9	49.1	40.7	35.5
	5%	DB	40.8	39.2	45.0	58.1	70.6	74.8	78.1	75.7	70.5	56.3	43.8	37.8
		MCWB	35.6	33.9	37.3	43.8	51.5	57.3	62.4	60.7	56.3	45.8	37.7	33.0
	10%	DB	36.9	35.0	41.4	53.4	66.8	71.7	75.0	72.5	65.3	52.7	39.9	32.9
		MCWB	32.8	31.1	35.0	41.4	49.8	56.0	60.8	59.4	53.1	43.7	34.7	29.2
	0.4%	WB	40.0	38.7	43.7	51.2	58.1	63.9	69.8	68.3	62.8	55.4	45.0	39.5
		MCDB	46.7	44.9	51.9	69.0	75.1	75.6	80.6	79.1	77.0	68.9	54.4	45.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.6	36.3	40.3	47.6	55.3	61.1	66.3	65.3	59.8	50.5	41.2	36.1
		MCDB	42.9	41.8	47.7	62.4	70.1	73.4	77.3	76.0	72.9	60.3	47.5	40.9
	5%	WB	35.7	34.1	37.7	44.4	53.1	59.2	64.2	62.7	57.0	47.0	38.2	32.9
		MCDB	40.5	39.0	44.7	56.3	67.0	71.6	75.0	72.4	68.5	54.0	43.2	37.6
	10%	WB	32.9	31.1	35.3	42.0	51.1	57.5	62.1	60.6	54.4	44.6	35.3	29.3
		MCDB	36.8	35.0	41.2	52.2	64.2	68.4	72.0	70.0	63.3	51.3	39.3	32.7
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.4	21.6	21.3	24.4	22.1	22.0	22.4	22.6	18.7	17.0	18.3
		MCDBR	18.9	23.4	23.8	31.1	32.4	30.0	28.9	29.8	31.8	26.4	19.6	22.3
		MCWBR	14.2	17.6	15.8	16.7	14.9	14.1	14.4	15.4	16.6	15.9	13.3	18.0
		MCDBR	18.2	22.5	23.0	29.6	29.0	25.3	26.9	27.5	30.7	25.4	18.9	22.6
	5% WB	MCWBR	14.1	17.1	15.3	16.5	13.9	12.5	14.6	15.8	16.6	16.1	13.6	18.8
(40) Clear-Sky Solar Irradiance	taub		0.240	0.252	0.286	0.332	0.347	0.352	0.354	0.350	0.313	0.285	0.252	0.232
	taud		2.350	2.405	2.406	2.389	2.400	2.406	2.433	2.419	2.535	2.576	2.448	2.283
	Ebn at Noon		231	272	285	283	284	283	280	274	272	257	226	203
	Edh at Noon		16	22	28	33	34	35	33	31	24	18	14	13
(44) All-Sky Solar Radiation	RadAvg		246	564	1053	1491	1766	1908	1871	1533	1041	565	259	174
	RadStd		23	45	85	96	129	146	124	122	89	66	20	16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A

Nomenclature: See separate page